

Practice Questions

- Cloud Migration – Benefits & Risks

Key Points:

Benefits: lower IT cost, scalability, auto-updates, centralised access.

Risks: downtime, vendor lock-in, privacy concerns, compliance risk.

- R&D Cloud Migration (Pharma Company)

Key Points:

Benefits: global collaboration, secure storage, compute power.

Risks: IP theft, data residency issues, cyber-espionage risk.

- Virtual Desktops for Financial Services

Key Points:

Centralised control, enhanced compliance, reduced data leakage, uniform security.

- Virtualization for EdTech Growth

Key Points:

Auto-scaling, workload isolation, low-cost scaling, improved uptime.

- Data Mining for Delivery Delays

Key Points:

Clustering, regression, route heatmaps, time-series analysis.

- Pattern Detection in Customer Complaints

Key Points:

Text mining, sentiment analysis, geo-mapping, classification.

- IaaS vs SaaS for CRM

Key Points:

IaaS: more control, customization.

SaaS: quick deployment, low maintenance, lower cost.

- PaaS vs SaaS for Rapid App Development

Key Points:

PaaS: flexible development, coding environment.

SaaS: ready-to-use, minimal setup.

- Omni-Channel Strategy (Bookstore / Fashion Brand)

Key Points:

Unified inventory, online ordering + in-store pickup, personalization, app integration.

- Digital Goods – Characteristics & Advantages

Key Points:

Zero marginal cost, instant delivery, scalable globally.

- Governance Frameworks for Non-profits

Key Points:

COBIT (controls), ITIL (service management), ISO 27001 (security).

Improves transparency, accountability, compliance.

- Compliance Challenges in Large Corporations

Key Points:

Need centralised dashboards, periodic audits, policy automation, documentation.

- Ethical Use of Data in Fintech

Key Points:

Consent management, encryption, audit logs, transparency, minimal collection.

- Biometric Data Governance

Key Points:

Encrypt biometrics, restrict access, comply with privacy laws, regular audits.

- Cybersecurity Threats to Hospitals

Key Points:

Ransomware, phishing, insider misuse.

Impact: delayed treatment, data loss, safety risks.

- Ransomware in Public Hospitals

Key Points:

Service disruption, halted emergency operations, reputational damage.

- Risk Assessment for Patient Portal

Key Points:

Identify threats → analyse impact → apply controls (MFA, encryption, monitoring).

- Risk Management for Citizen Portal

Key Points:

NIST/ISO cycle: identify → assess → mitigate → monitor → review.

Case Study Summary – Big Data / MegaMart

A large retail chain collects massive amounts of data from stores, warehouses, apps, and loyalty systems. However, the data is under-used due to silos and manual reporting. The company considers Big Data analytics and BI dashboards to improve forecasting, customer insights, and operations.

Case Study Summary – IT Governance / Global Bank

A multinational bank fails an IT audit due to weak controls, inconsistent processes, and outdated systems. Leadership plans to adopt governance frameworks like COBIT and ITIL to strengthen compliance, risk management, and IT operations.

Case Study Summary – Privacy & Emerging Tech / Govt AI Platform

A government launches an AI-driven citizen platform using biometrics and cloud systems. Concerns arise around privacy, cybersecurity, and bias. Proper governance, ethics, and data protection measures are needed.

CASE STUDY 1 – BIG DATA / MEGAMART

- Explain the relevance of Big Data's 5Vs in this case.

Key Points: Volume (millions of transactions), Velocity (real-time updates), Variety (structured + unstructured), Veracity (quality checks needed), Value (decision insights).

- Identify two ways forecasting helps MegaMart.

Key Points: Reduces stockouts, improves procurement planning.

- How does BI support decision-making?

Key Points: Dashboards, KPIs, drill-downs, automated reporting.

- Suggest ethical guidelines for customer data.

Key Points: Consent, anonymisation, minimal collection, encryption.

CASE STUDY 2 – IT GOVERNANCE / GLOBAL BANK

- What governance failures do you see?

Key Points: Outdated controls, poor access management, no standard processes, weak audit trails.

- How does COBIT help?

Key Points: Aligns IT with business goals, improves accountability, defines controls.

- Why is ITIL important here?

Key Points: Controls incidents, change management, prevents recurring issues.

- Suggest compliance KPIs.

Key Points: Patch compliance %, audit closure rate, high-risk incident count.

CASE STUDY 3 – PRIVACY & AI ETHICS / GOVT PLATFORM

- Identify the major privacy concerns.

Key Points: Over-collection of data, biometric risks, unclear sharing policies.

- Explain key cybersecurity gaps.

Key Points: Weak authentication, poor encryption, vulnerable APIs.

- What are the AI ethics issues?

Key Points: Bias, unfair decisions, lack of transparency.

- What governance measures are required?

Key Points: Explainable AI, human oversight, role-based access, encrypted storage.